

## Field Testing



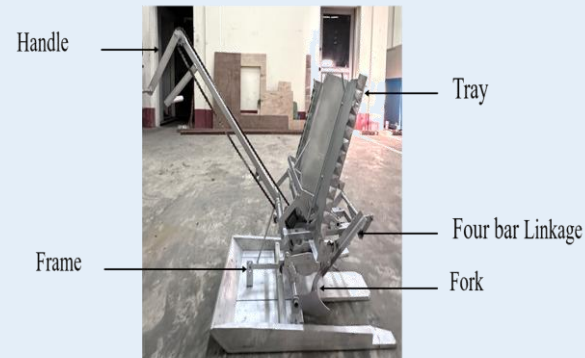
- ❖ Prepared and leveled the field bed for a smooth soil surface.
- ❖ Placed seedling trays on the bed to grow uniform seedling mats.
- ❖ Turned the hand-crank continuously to actuate the four-bar linkage mechanism.
- ❖ Observed planting speed, planting depth and machine behavior.



## Testing Focus

- ❖ Planting speed
- ❖ Planting depth accuracy
- ❖ Practical operation in paddy soil

## Experimental Results

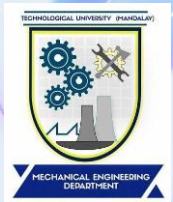


| Performance                | Value                      |
|----------------------------|----------------------------|
| Area per hill              | 0.036 m <sup>2</sup> /hill |
| Theoretical Field Capacity | 129.6 m <sup>2</sup> /hr   |
| Theoretical Operating Time | 31 hrs/acre                |
| Field Efficiency           | 70%                        |
| Actual Operating Time      | 44 hrs/acre                |

## Performance Summary

- ❖ The reported field efficiency is 70%. Theoretical field capacity is 129.6 m<sup>2</sup>/hr, with theoretical and actual operating times of 31 and 44 hrs/acre, respectively.

Technological University (Mandalay)  
Department of Mechanical Engineering



# HAND-CRANKED RICE TRANSPLANTER



Hand-Cranked Rice Transplanter

## Introduction

- ❖ Rice is one of the most important food crops in Myanmar. Traditional rice transplanting is time-consuming, labor-intensive and physically demanding for farmers.
- ❖ The hand-cranked rice transplanter is a simple and low-cost agricultural machine that converts rotary hand-crank motion into reciprocating planting motion through a linkage mechanism.



## Purpose

- ❖ The main purpose of the project is to compare the time, depth and space of plant of manual rice planting and hand-cranked rice transplanter which can be used for farmers to transplant rice planting easily onto the paddy field.
- ❖ Solving labor shortage issues during the peak transplanting season.
- ❖ The design is to significantly lower operational costs and save planting time.

## Methodology

1. Requirement
  - ✓ Fixed row spacing and planting depth
  - ✓ Low hand-crank effort for the operator
2. Design and Kinematics
  - ✓ Four-bar linkage for smooth seedling picking
  - ✓ Lightweight frame for soft paddy soil
3. Fabrication
  - ✓ Durable, local and low-cost materials
  - ✓ Corrosion-resistant body structure
4. Field Testing
  - ✓ Evaluation of planting speed and depth accuracy

## Operations

- ✓ Prepare the rice seedlings for transplanting.
- ✓ Place the seedlings in the seedling tray.
- ✓ Position the transplanter at the starting point of the field.
- ✓ Adjust the planting depth and required spacing.
- ✓ Rotate the hand-crank continuously.
- ✓ The planting finger picks up and places seedlings into the soil.



## Main Components of Hand-Cranked Rice Transplanter



## Application Area

- ✓ Intended for rice transplanting in paddy fields, especially for small-scale farmers in Myanmar.

## Field Test Location

- ✓ Agricultural field along Mandalay-Myintge Road.

